INCH-POUND
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5 August 2011
SUPERSEDING
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PERFORMANCE SPECIFICATION

CLAMP, REPAIR, PIPE: WITH FUEL-RESISTANT GASKET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of clamps one for split application and the other for pinhole application over damaged petroleum pipelines.

1.2 Classification. Clamps are of the following types, and sizes, as specified (see 6.2):

1.2.1 Types. The type of repair clamps are as follows:

Type I - Split application repair clamp

Type II - Pinhole application repair clamp for bullet-hole leaks

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AMSC N/A FSC 4730

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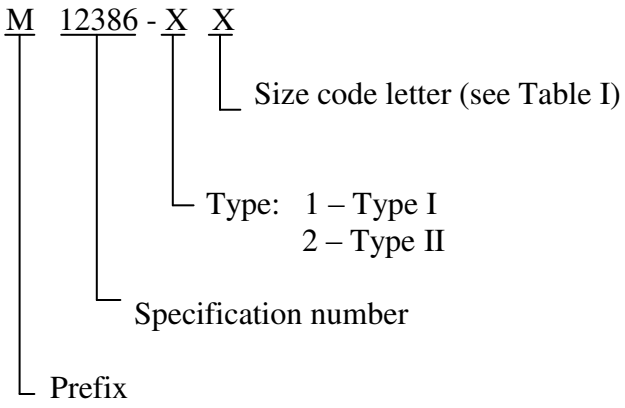
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1.2.2 Sizes. The size of repair clamps are of the following, as specified in Table I.

TABLE I. Repair Clamp Sizes.

Size	Clamp Size in Inches Nominal
A	4
B	6
C	8
D	12

1.3 Part or identification number. The specification part number of the clamps covered by this specification will be designated in the following form (see 6.2):



EX: M12386-1C – Where M12386 indicates prefix and the specification number, -1 indicates Type I clamp and C indicates 8 inch clamp size.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-595/33446 - Tan 686A Camo

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-889 - Dissimilar Metals
MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product
MIL-STD-130 - Identification Marking of US Military Property

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A194/A194M - Specification for Carbon and Alloy Steel Nuts for Bolts for high-Pressure and high Temperature Service

ASTM B633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

(Copies of these documents are available from www.astm.org or ASTM International, P.O. Box C700, West Conshohocken, PA 19428-2959.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

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(Copies of these documents are available from www.asq.org or American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Clamps for split application shall have two identical semicircular half shells with flange that can be bolted together and clamps for pin-hole application shall have pilot pin with cone and guide and draw-bolt at the opposite side and as specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor/manufacturer and shall be subject to all provisions of this specification. The used, rebuilt, or remanufactured components, pieces or parts shall not be permitted.

3.3.1 Material deterioration and control. The pipe shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the clamps may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4 Design considerations for Type I clamps: Type I repair clamps shall be provided with following components.

3.4.1 Body or outer shell. Outer shell of the repair clamp for split application shall be designed to with stand an assault pipe under at least 900 psig and comprised of two identical semicircular halves. Length of split in pipe could be as long as 3 inches. Material for body or outer shell shall be selected by the manufacturer of split clamp. Curvature of the shell shall be

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such that it can be clamped over the specified pipe size. Each shell half shall have flange with bolt holes on both sides of the cylindrical halves such that the two halves can be bolted together.

3.4.2 Gasket or packing. Gasket or packing material shall be suitable for use with petroleum based fuels such as JP-8, JetA-1, diesel, kerosene or gasoline. Gasket or packing shall be factory installed firmly in the body or outer shell to prevent leakage of fluids. Gasket or packing material shall not deteriorate while the repair clamps are tested with water.

3.4.3 Bolts, nuts and washers. The contractor/manufacturer shall select the proper type, size of bolts, nuts, flat washers and lock washers. Cadmium plating is not acceptable. If bolts, nuts washers are galvanized, thickness of zinc coating shall be Fe/Zn 12 per ASTM B633. The coating shall not use ozone depleting substances, such as 1,1,1 Trichloroethane. Alternative methods or solvents shall be used in place of ozone depleting substances.

3.5 Design considerations for Type II clamps: Type II repair clamps for pin-hole application shall be provided with following components.

3.5.1 Band . The outer band of the pin-hole repair shall be designed such a way that when draw-bolt is tightened it can withstand operating pressure of 900 psig. It shall also be capable of housing a pilot-pin cone or a force screw which can be tightened over the leaking hole of the assault pipe. Material for the hose clamp type band shall be selected by the manufacturer although corrosion resistant material such as type 304 stainless steel is preferred.

3.5.2 Pilot-pin cone and force screw. Pilot-pin cone and force screw shall be designed by the manufacturer of the repair clamp. Force screw and the cone shall be capable directly positioned over the leak without too much effort. Cone can be made of Buna-N or other material suitable resistant to JP-8 and petroleum based fuels. If necessary, a thrust washer can be placed between force screw and the rubber cone. Material for the force-screw, cone and thrust washer shall be selected by the contractor or manufacturer of the clamp.

3.5.3 Draw-bolt. Bolt sizes and end shapes suitable for the pin-hole clamp shall be determined by the contractor. Draw-bolt shall be configured and fastened to the band such a way that band can be easily tightened on the pipe by rotating the nut of the draw-bolt. Suitable material for the draw-bolt shall be chosen by the manufacturer so that under maximum operating pressure there is no deformation or twisting of band or any other part of this clamp.

3.6 Nuts. Nuts for the stud bolts of the Type I clamps shall conform to ASTM A194/A194M, Grade 2M, heavy hex series.

3.7 Washers. Washers shall be furnished with all stud bolts (type I clamps), and the pin-hole clamps shall match bolt sizes, to be selected by the manufacturer of the clamps. Washers shall be galvanized with thickness of Fe/Zn-12 per ASTM B633. The coating shall not use ozone depleting substances, such as 1,1,1 Trichloroethane. Alternative methods or solvents shall be used in place of ozone depleting substances.

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3.8 Hydrostatic pressure. Type I clamps shall not leak when subjected to the hydrostatic pressure specified for each size clamp, as follows:

Clamp (nominal pipe size)	Pressure pounds per square inch (psig)
4 inches	1350
6 inches	1350
8 inches	1350
12 inches	1350

3.9 Identification marking. Clamps shall be identified by rolling, stamping, or stenciling in accordance with MIL-STD-130.

3.10 Treatment and painting. Unless otherwise specified (see 6.2), all surfaces of the clamps, normally painted internally and externally shall be cleaned, treated and painted in accordance with manufacturer's standard practice and the top coat color outer exposed surfaces shall be Tan 686A Camo, #33446 of FED-STD-595.

3.11 Workmanship.

All external surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety. Surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 Examination. Clamps shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.2.2 Tests. Clamps shall be tested as specified in 4.4.2. Failure of any test shall be cause for rejection.

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4.3 Conformance inspection.

4.3.1 Lot. For the purpose of inspection, a lot shall consist of all types, and pipe sizes of clamps offered for delivery at one time.

4.3.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-1916 or ANSI/ASQC Z1.4.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined as specified in 4.5.1.

4.3.4 Tests. Samples selected in accordance with 4.3.2 shall be tested as specified in 4.5.2. Inspection level shall be S-2. AQL shall be 1.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. Clamps shall be examined as specified herein for the following defects:

101. Materials not as specified (see 3.3 through 3.7).
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the repair clamps (see 3.3).
105. Dimensions or components not in accordance with applicable specification sheet (see 3.1).
106. Components missing. Each missing component shall constitute one failure (see 3.1).
107. Clamps not of types, and sizes specified (see 3.1).
108. Workmanship not as specified (see 3.13).
109. Treatment and painting not as specified (see 3.10).
110. Paint color not as specified (see 3.10).
111. Welding not as specified (see 3.13.3).
112. Contractor does not have documentation available for materials, material finishes or treatments (see 3.3.3).
113. Identification marking missing, incomplete, or illegible (see 3.9).

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4.4.2 Tests.

4.4.2.1 Hydrostatic tests. Clamps shall be subjected to the following hydrostatic tests as applicable. The test fluid shall be water.

4.4.2.1.1 Type I (split application). Test as follows: (a) In a pipe of the correct size for the clamp, prepare a slit 2 inches long and 0.063 inch wide. (b) Apply clamp with gasket over the slit. (c) Slowly raise the water pressure to the applicable hydrostatic test pressure in steps of 50 psi from 1000 to 1350 psig or as specified in 3.8 and hold for 5 minutes. Evidence of leakage shall constitute failure of this test.

4.4.2.1.2 Type II (pinhole application). Test as follows: (a) In proper size pipe, prepare an artificial leak 0.25 inch in diameter. (b) Apply clamp with gasket over the leaking area. (c) Raise the water pressure to the applicable hydrostatic pressure as specified in 3.8 and hold for 2 minutes 30 seconds. Evidence of leakage shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Clamps are intended to be used for emergency repair of pinholes and/or splits in pipelines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Definitive part number of clamp, including type, and size required (see 1.3).
c. If required, the specific issue of individual documents referenced (see 2.2.1 and 2.3)
d. When first article is required, the time frame required for submission of first article model, and number of clamps required (see 3.2).

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e. Treatment, painting, or color if other than specified (see 3.10).
f. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Subject term (key word) listing.

Pinhole application
Split application

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Air Force - 99

Review activities:
Navy - MC
DLA - CC

Preparing activity:
Army - AT

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